

ANAND BHALERAO

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Experience

FIS Global

Pune, India

Software Engineer 2

Jan 2026 - Present

- Designed and shipped **real-time session revocation** across multi-tenant systems serving 40+ banking clients as the primary defense against compromised authenticated sessions with zero downtime, addressing a **\$2M fraud incident**; built atomic invalidation using Spring + PostgreSQL.
- Architected **field-level AES encryption** using a custom MyBatis interceptor, adopted by 6 engineering teams to transparently rewrite 80% of application queries while **securing 1M+ PII records** with only **0.2ms overhead**.
- Designed a reusable AES key rotation framework across 10+ environments; built a resumable, chunked migration pipeline with pre-flight validation (config, version, readiness checks), version tracking, and idempotent re-encryption — **migrating 1M+ records** with full rollback capability and **zero unsafe rotations**.
- Implemented blind indexing with clustered indexes for encrypted fields, reducing database lookup time from **47s to 60ms** while maintaining field-level encryption.
- Designed a **lazy credential migration** framework to transparently upgrade hundreds of thousands of password and challenge-answer hashes from SHA-256 to SHA-512 during successful authentication, **enabling incremental migration** without bulk reprocessing or user disruption.

Software Engineer 1

Jul 2024 - Jan 2026

- Resolved **burst-traffic row-lock contention** by replacing a single-row sequence table with block-based cached allocation, **eliminating request timeouts under load**.
- Replaced high-overhead in-memory pagination by implementing database-level pagination, sorting, and filtering; reduced data fetched per request from **90K rows to 10 rows**, improving latency for **100K+ datasets**.
- Led the end-to-end Zelle SSO integration for **12+ banking clients**, implementing user-state reconciliation that synchronized external Zelle records with the bank's source of truth before SSO, alongside automated lifecycle workflows.
- Redesigned multi-entity session handling, eliminating recurring user lockouts and enabling seamless account switching without redundant re-authentication.
- Integrated Outseer behavioral biometrics into the authentication layer, enabling **risk-based step-up authentication** from client-side device signals.

Software Development Engineer Intern

Jan 2024 - Mar 2024

- Built end-to-end Electronic File Delivery, delivering 8 Angular screens and 6 Spring REST APIs.
- Engineered an interstitial for mandatory contact updates, with admin-controlled retry logic to ensure high compliance.

Projects

Distributed UPI Payment Simulator

Demo Video

Sarvatra Technologies

- Architected a **distributed UPI simulator** modeled on NPCI switch architecture using Spring Boot, Redis & Kafka.
- Built fault-tolerant payment processing with idempotency, Saga-based distributed transactions, and Redis-backed Bloom filters **reducing database lookups by 92%**.
- Instrumented end-to-end observability (OpenTelemetry + Prometheus) and load tested up to **1,800 TPS**, achieving **85ms p95 and 140ms p99** latency under sustained load.
- Containerized microservices with Docker and **deployed multi-replica workloads on Kubernetes** (k3s) running on AWS EC2; built GitHub Actions CI/CD pipelines for automated testing, image publishing, and zero-downtime rolling.
- Implemented secure authentication using JWT for session management and FIDO2/WebAuthn for passwordless authentication with public-key cryptography.

Education

Pune Institute of Computer Technology

Jun 2020 - Jun 2024

B.E in Computer Engineering (CGPA - 8.94)

Pune, Maharashtra

Achievements

- 1100+ problems** solved across LeetCode (peak 1763), CodeChef (AIR 145, 496 in Starters), and Codeforces.
- Finalist in Smart India Hackathon '22.

Technical Skills

Languages: Java, Kotlin, Typescript

Application Development: Spring Boot, MyBatis, Hibernate, JPA, Angular, Compose Multiplatform

Databases & Cache: PostgreSQL, Oracle, MySQL, Redis

Infrastructure & Observability: Kafka, OpenTelemetry, Prometheus